THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 17	The common difference is 3, so add 3 to the last term.
2	**(a)** acute	Compare 30° with 90° and 180°.
3	**(a)** 8531	Write the digits in decreasing order.
4	**(a)** 30	Multiply the symbols by the value of one symbol.
5	**(c)** 13	13 has only two factors, 1 and 13.
6	**(b)** 15	The 8th odd number is $2 \times 8 - 1$.
7	**(c)** 60	The hour hand turns 30° in 1 hour.
8	**(d)** 1358	Write the digits in increasing order (a non-zero digit first).
9	**(c)** 35	Multiply the height by the scale.
10	**(a)** 27	27 has more than two factors.
11	**(b)** 14	The 7th even number is 2×7.
12	**(a)** 90	A full turn is 360°.
13	**(b)** 454	A palindrome reads the same forwards and backwards.
14	**(d)** 1 bundle of five and 2 single strokes	Every fifth stroke crosses the four before it, making a bundle of five.
15	**(d)** 6	Factors: 1, 2, 3, 4, 6, 12.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Common difference $= 5$ (1); Next three: $24, 29, 34$ (1).	1+1
17	It is acute (1); Compare with the right angle 90° (and the straight angle 180°) (1).	1+1
18	Largest 9752 (1); Smallest 2579 (1).	1+1
19	Each symbol is 10 students (1); $6 \times 10 = 60$ students (1).	1+1
20	Find pairs whose product is 16 (1); Factors: 1, 2, 4, 8, 16 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Common difference $= 2$ (1); Term 8 $=$ first term $+7\times$ difference $= 3 + 7 \times 2$ (1); 17 (1).	1+1+1
22	$quarter$ of 360° (1); $= 90^\circ$ (1); Type: right (1).	1+1+1
23	Largest $= 741$, smallest $= 147$ (1); Sum $= 888$ (1); Difference $= 594$ (1).	1+1+1
24	Add all the numbers (1); Total $= 15 + 35 + 10 + 15 = 75$ (1); Hence the total is 75 (1).	1+1+1
	OR: Divide by primes in turn (1); $36 = 2 \times 2 \times 3 \times 3$ (1); Only prime factors appear (1).	1+1+1
25	$1 + 3 + 5 + 7 + 9 = 25$ (1); $25 = 5 \times 5$ is a square number (2).	1+2
	OR: Each hour mark is 30° apart (1); Angle $= 1 \times 30 = 30^\circ$ (1); Type: acute (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	First 6 terms: $2, 5, 8, 11, 14, 17$ (1); Term 7 is 20 and term 8 is 23 (2); Yes, 23 is the 8th term (2).	1+2+2
	OR: 30° is acute (1); $360 \div 30 = 12$ (1); 12 turns make a full turn (1); Check: product is 360° (2).	1+1+1+2
27	Total $= 1245 + 1378 = 2623$ (1); Difference $= 1378 - 1245 = 133$ (2); Total strength is odd (2).	1+2+2
	OR: A: $4 \times 10 = 40$ (1); B: $6 \times 10 = 60$ (1); C: $5 \times 10 = 50$ (1); Total $= 150$ (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 36	$6 \times 6 = 36$.
2	**(b)** 3	Count every pair of points.
3	**(b)** 1000	Add 1.
4	**(c)** Grapes	Look for the largest number.
5	**(b)** 42	7×6
6	**(a)** 10	$\frac{4\times5}{2} = 10$.
7	**(d)** two end points	Definition
8	**(c)** 999	Subtract 1.
9	**(a)** Football	Look for the smallest number.
10	**(b)** 363	Divisibility test of 3.
11	**(d)** 8	$2 \times 2 \times 2 = 8$.
12	**(b)** 30	In 1 minute the minute hand turns 6°.
13	**(a)** 400	Place value = digit × value of its place.
14	**(c)** 2	Count (tally) the number of times.
15	**(c)** 10	Add the primes 2, 3, 5, 7, ... (1 is not prime).

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Squares are $1 \times 1, 2 \times 2, \ldots$ (1); $1, 4, 9, 16, 25$ (1).	1+1
17	It turns 30° per hour (1); $3 \times 30 = 90^\circ$ (1).	1+1
18	Reverse of 636 is 636 (1); Yes, it is a palindrome (1).	1+1
19	Draw strokes; every fifth stroke crosses four (1); 9 = 1 bundle of five and 4 single strokes (1).	1+1
20	Multiply 6 by 1, 2, 3, 4, 5 (1); 6, 12, 18, 24, 30 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Reverse = 21 (1); Sum = $12 + 21 = 33$ (1); Yes, palindrome (1).	1+1+1
22	Count each value (1); 1: 2; 2: 5; 3: 4; 4: 4 (1); Total = 15 (1).	1+1+1
23	By 2: no (last digit 5) (1); By 3: yes (digit sum 9) (1); By 5: yes (1).	1+1+1
24	$1 + 2 + \cdots + 9$ (1); $= 45$ (2).	1+2
	OR: It turns 6° per minute (1); $8 \times 6 = 48^\circ$ (1); Less than 90° (1).	1+1+1
25	Middle row: $3 + 5 = 8$ and $5 + 2 = 7$ (1); Top number = $8 + 7$ (1); $= 15$ (1).	1+1+1
	OR: Most: Tennis (50) (1); Least: Football (25) (1); Difference = 25 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	$12 = 2 \times 2 \times 3$ (1); $18 = 2 \times 3 \times 3$ (2); Common prime factors: 2, 3 (2).	1+2+2
	OR: $1 + 3 + 5 + 7 = 16$ (1); $16 = 4 \times 4$ (2); Sum of first 5 odd numbers = $5 \times 5 = 25$ (2).	1+2+2
27	Segments: $4 \times 3 \div 2 = 6$ (1); A can be paired with each of the other 3 points (1); 3 segments have A as an end point (1); A segment has two end points and a fixed length (2).	1+1+1+2
	OR: Greatest = 8642 (1); Smallest = 2468 (1); Difference = 6174 (1); Even (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 16	The sum of the first n odd numbers is $n \times n$.
2	**(a)** right	Compare 90° with 90° and 180°.
3	**(c)** even	Rule
4	**(a)** data	Definition
5	**(a)** 2	Definition
6	**(c)** 25	The common difference is 4, so add 4 to the last term.
7	**(c)** 90	The hour hand turns 30° in 1 hour.
8	**(c)** 8	Count places from the right: ones, tens, hundreds, thousands.
9	**(d)** 70	Multiply the symbols by the value of one symbol.
10	**(d)** 29	29 has only two factors, 1 and 29.
11	**(c)** 23	The 12th odd number is $2 \times 12 - 1$.
12	**(b)** 180	A full turn is 360°.
13	**(a)** 9642	Write the digits in decreasing order.
14	**(d)** 60	Multiply the height by the scale.
15	**(d)** 49	49 has more than two factors.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Add 1, then 2, then 3, ... one at a time (1); $1, 3, 6, 10, 15$ (1).	1+1
17	Each pair of points gives a segment (1); 7 points give 21 segments (1).	1+1
18	Successor $= 4999 + 1 = 5000$ (1); Predecessor $= 4999 - 1 = 4998$ (1).	1+1
19	Multiply: 4×10 (1); 40 students (1).	1+1
20	Factors of 31: 1, 31 (1); Prime: only two factors (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Test each number for factors (1); 11, 13, 17, 19, 23, 29 (2).	1+2
22	Common difference $= 3$ (1); Term 7 $=$ first term $+6\times$ difference $= 5 + 6 \times 3$ (1); 23 (1).	1+1+1
23	$half$ of 360° (1); $= 180^\circ$ (1); Type: straight (1).	1+1+1
24	Largest $= 953$, smallest $= 359$ (1); Sum $= 1312$ (1); Difference $= 594$ (1).	1+1+1
	OR: Add all the numbers (1); Total $= 35 + 15 + 25 + 10 = 85$ (1); Hence the total is 85 (1).	1+1+1
25	Divide by primes in turn (1); $48 = 2 \times 2 \times 2 \times 2 \times 3$ (1); Only prime factors appear (1).	1+1+1
	OR: $1 + 3 + 5 + 7 + 9 + 11 = 36$ (1); $36 = 6 \times 6$ is a square number (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Monday 30 (1); Tuesday 50 (1); Wednesday 40 (1); Total 120 books (2).	1+1+1+2
	OR: Factors: 1, 2, 4, 7, 14, 28 (1); Count $= 6$ (1); Sum of all factors $= 56$ (1); Even (2).	1+1+1+2
27	First 6 terms: $4, 9, 14, 19, 24, 29$ (1); Term 7 is 34 and term 8 is 39 (2); Yes, 39 is the 8th term (2).	1+2+2
	OR: 45° is acute (1); $360 \div 45 = 8$ (1); 8 turns make a full turn (1); Check: product is 360° (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.